

Companion to Bird & de Moor, Algebra of Programming

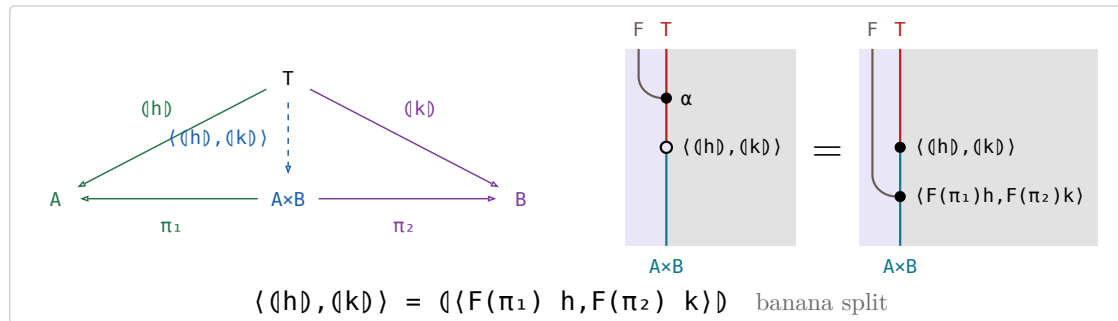
§3. Applications

§3.1. Banana-split

(3.1a)

definition	type	note
$\text{listr } A ::= \text{nil} \mid \text{cons}(A, \text{listr } A)$	$\mathcal{A} \rightarrow \mathcal{A}$	The cons-lists over A , the datatype every row below folds.
$\text{sum} = \llbracket [\text{zero}, \text{plus}] \rrbracket$	$[A] \rightarrow A$	$\text{plus}(a, b) = a + b$.
$\text{length} = \llbracket [\text{zero}, \pi_2 \text{ succ}] \rrbracket$	$[A] \rightarrow \text{Nat}$	π_2 drops the head and keeps the count of the tail, succ adds one for the head.
$\text{average} \triangleq (\text{sum}, \text{length}) \text{ div}$	$[\text{Nat}] \rightarrow \text{Nat}$	$\text{div}(m, n) = m/n$, with $\text{div}(0, 0) = 0$ so average is total. Traverses the list twice.
banana-split law $\langle \llbracket h \rrbracket, \llbracket k \rrbracket \rangle = \llbracket (F(\pi_1) h, F(\pi_2) k) \rrbracket$	$T \rightarrow A \times B$	Any fork of folds is a single fold, hence one traversal — F the base functor.
what it reduces to $\alpha \langle \llbracket h \rrbracket, \llbracket k \rrbracket \rangle = F(\langle \llbracket h \rrbracket, \llbracket k \rrbracket \rangle) \langle F(\pi_1) h, F(\pi_2) k \rangle$	$FT \rightarrow A \times B$	All that $\triangleq$ reduce leaves to check: the fork satisfies the defining equation.
the instance $\langle \text{sum}, \text{length} \rangle = \llbracket [\text{zeros}, \text{pluss}] \rrbracket$	$[\text{Nat}] \rightarrow \text{Nat} \times \text{Nat}$	$\text{pluss}(a, (b, n)) = (a + b, n + 1)$, so average runs in one pass.
$\text{preds } n = [n, n-1, \dots, 1]$	$\text{Nat} \rightarrow [\text{Nat}]$	apply the earlier special case to write preds as $\llbracket k \rrbracket \pi_1$.

(3.1b)



§3.1.1. Fokkinga's mutual recursion theorem

- $F\text{-Alg}(\mathcal{A})$ — the algebras and their homomorphisms — has binary products whenever $\mathcal{A}$ does, and the forgetful $U : F\text{-Alg}(\mathcal{A}) \rightarrow \mathcal{A}$ **creates** them: the product of $h : FA \rightarrow A$ and $k : FB \rightarrow B$ is carried by $A \times B$, with structure $\langle F(\pi_1)h, F(\pi_2)k \rangle : F(A \times B) \rightarrow A \times B$.
- π_1 and π_2 are homomorphisms out of it, and $\langle p, q \rangle : X \rightarrow A \times B$ is a homomorphism **iff** p and q both are — substitute the structure and compare the two forks component by component.
- The banana-split law of (3.1a) IS that product's universal property read at the initial algebra: $\langle h \rangle$ and $\langle k \rangle$ are the unique homomorphisms to (A, h) and (B, k) , so their fork is the unique homomorphism into the product, which is $\langle \langle F(\pi_1)h, F(\pi_2)k \rangle \rangle$.
- Fokkinga's theorem is **strictly more general** and is not that product: in mutual recursion the two arrows leave $F(A \times B)$, not FA and FB , so each sees BOTH components. The product is the case where they factor as $F(\pi_1)h$ and $F(\pi_2)k$. What it says: **any algebra on a product carrier is folded by a fork**.

(3.1.1a)



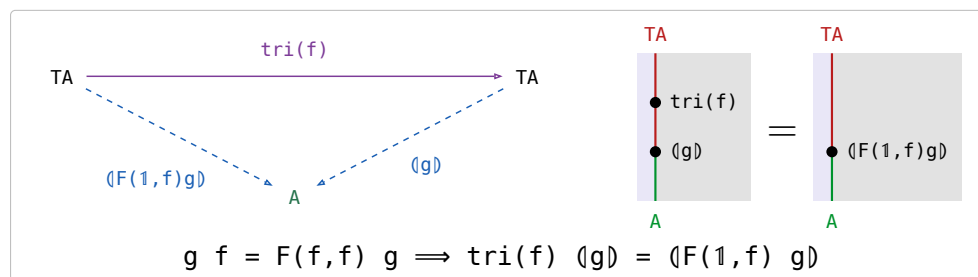
§3.2. Ruby triangles and Horner's rule

(3.2a)

stage	definition
informally	$\text{tri}(f)[a_0, a_1, \dots, a_i, \dots, a_n] = [a_0, f(a_1), \dots, f^i(a_i), \dots, f^n(a_n)]$
for cons-lists	$\text{tri}(f) = \langle [\text{nil}, (1 \times \text{listr}(f)) \text{ cons}] \rangle$
with the base functor named	$F(A, B) = 1 + A \times B$, $\alpha = [\text{nil}, \text{cons}]$, so $\text{tri}(f) = \langle F(1, \text{listr}(f)) \alpha \rangle$
abstractly	F a bifunctor with initial type (α, T) : $\text{tri}(f) = \langle F(1, T(f)) \alpha \rangle$

For the definition to make sense $f : A \rightarrow A$ is required, and then $\text{tri}(f) : TA \rightarrow TA$.

(3.2b)



Horner's rule is Horner's rule, generalised from cons-lists to any initial type. Fusion reduces it to $F(1, T(f))\alpha(g) = F(1, (g))F(1, f)g$.

§3.2.1. Depth of a tree

(3.2.1a)

the datatype	$\text{tree } A ::= \text{tip}(A) \mid \text{bin}(\text{tree } A, \text{tree } A)$, base functor $F(A,B)=A+B \times B$, initial type $([\text{tip}, \text{bin}], \text{tree})$
the fold	$\llbracket g, h \rrbracket$ is the unique f with $f(\text{tip}(a))=g(a)$ and $f(\text{bin}(x,y))=h(f(x), f(y))$
$\text{tree}(f)$	$\text{tree}(f) = \llbracket F(f, 1) \llbracket \text{tip}, \text{bin} \rrbracket \rrbracket$; pointwise $\text{tree}(f)(\text{tip}(a))=\text{tip}(f(a))$ and $\text{tree}(f)(\text{bin}(x,y))=\text{bin}(\text{tree}(f)(x), \text{tree}(f)(y))$
max	$\text{max} = \llbracket 1, \text{bmax} \rrbracket$, where $\text{bmax}(a,b)$ is the larger of a and b
depths	$\text{depths} = \text{tree}(\text{zero}) \text{ tri}(\text{succ})$ — replaces every tip by its depth in the tree, zero the constant function returning 0 and succ the successor
depth	$\text{depth} = \text{depths } \text{max}$

§3.3. The TeX problem

§3.4. Conditions and conditionals

§3.5. Concatenation and currying

If φ is natural, i.e. $G(h \times id) \cdot \varphi = \varphi \cdot (F(h) \times id)$, then $h \cdot G(f) \cdot \varphi = f \iff f = \text{apply} \cdot ((h \cdot G(\text{apply}) \cdot \varphi) \times id)$ Theorem 3.1
 The unique solution of a two-argument recursion $f(a, x) = h(G(f)(a, x))$ built from a natural φ is `apply` composed with a one-argument fold, which is how `cat` — not itself a fold — is derived from one.